TABLE B-3
RELATIVE PERMEABILITY μ_r OF SOME COMMON MATERIALS[a]

$\mu = \mu_r\mu_0$ and $\mu_0 = 4\pi \times 10^{-7}$ H/m.

Material	Relative Permeability, μ_r
Diamagnetic	
Bismuth	$0.99983 \approx 1$
Gold	$0.99996 \approx 1$
Mercury	$0.99997 \approx 1$
Silver	$0.99998 \approx 1$
Copper	$0.99999 \approx 1$
Water	$0.99999 \approx 1$
Paramagnetic	
Air	$1.000004 \approx 1$
Aluminum	$1.00002 \approx 1$
Tungsten	$1.00008 \approx 1$
Titanium	$1.0002 \approx 1$
Platinum	$1.0003 \approx 1$
Ferromagnetic (nonlinear)	
Cobalt	250
Nickel	600
Mild steel	2,000
Iron (pure)	4,000–5,000
Silicon iron	7,000
Mumetal	$\sim 100,000$
Purified iron	$\sim 200,000$

[a]These are typical values; actual values depend on material variety.

Note: Except for ferromagnetic materials, $\mu_r \approx 1$ for all dielectrics and conductors.

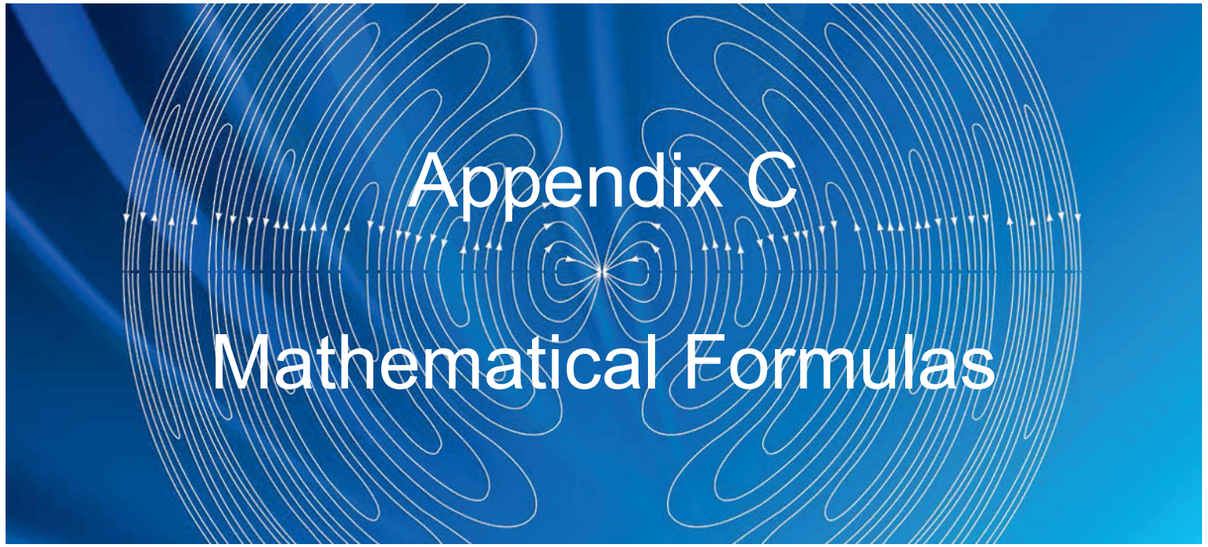
表 B-3 一些常见材料的相对磁导率 μ_r[a]

$\mu = \mu_r\mu_0$ 和 $\mu_0 = 4\pi \times 10^{-7}$ H/m。

材料	相对的渗透性，μ_r
抗磁性	
铋	$0.99983 \approx 1$
金	$0.99996 \approx 1$
汞	$0.99997 \approx 1$
银	$0.99998 \approx 1$
铜	$0.99999 \approx 1$
水	$0.99999 \approx 1$
顺磁性	
Air	$1.000004 \approx 1$
铝	$1.00002 \approx 1$
钨	$1.00008 \approx 1$
钛	$1.0002 \approx 1$
铂	$1.0003 \approx 1$
铁磁 （非线性）	
钴	250
镍	600
低碳钢	2,000
铁（纯）	4,000–5,000
硅铁	7,000
镍铁合金	$\sim 100,000$
精铁	$\sim 200,000$

[a]这些是典型值；实际值取决于材料的品种。

注：除铁磁材料外，$\mu_r \approx 1$ 适用于所有电介质和导体。



Appendix C
Mathematical Formulas

Trigonometric Relations

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$
$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$
$$2\sin x \sin y = \cos(x-y) - \cos(x+y)$$
$$2\sin x \cos y = \sin(x+y) + \sin(x-y)$$
$$2\cos x \cos y = \cos(x+y) + \cos(x-y)$$
$$\sin 2x = 2\sin x \cos x$$
$$\cos 2x = 1 - 2\sin^2 x$$
$$\sin x + \sin y = 2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)$$
$$\sin x - \sin y = 2\cos\left(\frac{x+y}{2}\right)\sin\left(\frac{x-y}{2}\right)$$
$$\cos x + \cos y = 2\cos\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)$$
$$\cos x - \cos y = -2\sin\left(\frac{x+y}{2}\right)\sin\left(\frac{x-y}{2}\right)$$
$$\cos(x \pm 90^\circ) = \mp \sin x$$
$$\cos(-x) = \cos x$$
$$\sin(x \pm 90^\circ) = \pm \cos x$$
$$\sin(-x) = -\sin x$$
$$e^{jx} = \cos x + j\sin x \qquad \text{(Euler's identity)}$$
$$\sin x = \frac{e^{jx} - e^{-jx}}{2j}$$
$$\cos x = \frac{e^{jx} + e^{-jx}}{2}$$

Approximations for Small Quantities

For $|x| \ll 1$,

$$(1 \pm x)^n \approx 1 \pm nx$$
$$(1 \pm x)^2 \approx 1 \pm 2x$$
$$\sqrt{1 \pm x} \approx 1 \pm \frac{x}{2}$$
$$\frac{1}{\sqrt{1 \pm x}} \approx 1 \mp \frac{x}{2}$$
$$e^x = 1 + x + \frac{x^2}{2!} + \cdots \approx 1 + x$$
$$\ln(1+x) \approx x$$
$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots \approx x$$
$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots \approx 1 - \frac{x^2}{2}$$
$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

附录 C

数学公式

三角恒等式

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$
$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$
$$2\sin x \sin y = \cos(x-y) - \cos(x+y)$$
$$2\sin x \cos y = \sin(x+y) + \sin(x-y)$$
$$2\cos x \cos y = \cos(x+y) + \cos(x-y)$$
$$\sin 2x = 2\sin x \cos x$$
$$\cos 2x = 1 - 2\sin^2 x$$
$$\sin x + \sin y = 2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)$$
$$\sin x - \sin y = 2\cos\left(\frac{x+y}{2}\right)\sin\left(\frac{x-y}{2}\right)$$
$$\cos x + \cos y = 2\cos\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)$$
$$\cos x - \cos y = -2\sin\left(\frac{x+y}{2}\right)\sin\left(\frac{x-y}{2}\right)$$
$$\cos(x \pm 90^\circ) = \mp \sin x$$
$$\cos(-x) = \cos x$$
$$\sin(x \pm 90^\circ) = \pm \cos x$$
$$\sin(-x) = -\sin x$$
$$e^{jx} = \cos x + j\sin x \qquad \text{（欧拉恒等式）}$$
$$\sin x = \frac{e^{jx} - e^{-jx}}{2j}$$
$$\cos x = \frac{e^{jx} + e^{-jx}}{2}$$

小量近似与级数展开

当 |x| 远小于1时：

$$(1 \pm x)^n \approx 1 \pm nx$$
$$(1 \pm x)^2 \approx 1 \pm 2x$$
$$\sqrt{1 \pm x} \approx 1 \pm \frac{x}{2}$$
$$\frac{1}{\sqrt{1 \pm x}} \approx 1 \mp \frac{x}{2}$$
$$e^x = 1 + x + \frac{x^2}{2!} + \cdots \approx 1 + x$$
$$\ln(1+x) \approx x$$
$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots \approx x$$
$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots \approx 1 - \frac{x^2}{2}$$
$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

Common Derivatives

$f(x)$	$\frac{df}{dx}$
x^a	ax^{a-1}
$\frac{1}{x^a}$	$-ax^{-(a+1)}$
$\sqrt{x}$	$\frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$
e^{ax}	ae^{ax}
b^x	$b^x \ln(b)$
$\ln(x)$	$\frac{1}{x}$
$\log_a(x)$	$\frac{1}{x\ln(a)}$
$\sin(ax)$	$a\cos(ax)$
$\cos(ax)$	$-a\sin(ax)$
$\tan(ax)$	$a\sec^2(ax)$

Common Integrals

$$\int x^n\,dx = \frac{1}{n+1}x^{n+1} + C$$

$$\int \frac{1}{x}\,dx = \ln|x| + C$$

$$\int x^{-n}\,dx = \frac{1}{-n+1}x^{-n+1} + C$$

$$\int \frac{1}{ax+b}\,dx = \frac{1}{a}\ln|ax+b| + C$$

$$\int \cos(ax)\,dx = \frac{1}{a}\sin(ax) + C$$

$$\int \sin(ax)\,dx = -\frac{1}{a}\cos(ax) + C$$

$$\int \tan(ax)\,dx = \frac{1}{a}\ln|\sec(ax)| + C$$

$$\int e^{ax}\,dx = \frac{1}{a}e^{ax} + C$$

$$\int b^x\,dx = \frac{b^x}{\ln(b)} + C$$

$$\int \ln x\,dx = x\,\ln(x) - x + C$$

$$\int e^{ax}\sin(bx)\,dx = \frac{e^{ax}}{a^2+b^2}(a\sin(bx) - b\cos(bx)) + C$$

$$\int e^{ax}\cos(bx)\,dx = \frac{e^{ax}}{a^2+b^2}(a\cos(bx) + b\sin(bx)) + C$$

$$\int xe^x\,dx = (x-1)e^x + C$$

Gradient, Divergence, Curl, and Laplacian Operators

Cartesian (Rectangular) Coordinates (x, y, z)

$$\nabla V = \hat{\mathbf{x}}\frac{\partial V}{\partial x} + \hat{\mathbf{y}}\frac{\partial V}{\partial y} + \hat{\mathbf{z}}\frac{\partial V}{\partial z}$$

$$\nabla\cdot\mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla\times\mathbf{A} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$= \hat{\mathbf{x}}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \hat{\mathbf{y}}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \hat{\mathbf{z}}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

Cylindrical Coordinates (r, ϕ, z)

$$\nabla V = \hat{\mathbf{r}}\frac{\partial V}{\partial r} + \hat{\boldsymbol{\phi}}\frac{1}{r}\frac{\partial V}{\partial \phi} + \hat{\mathbf{z}}\frac{\partial V}{\partial z}$$

$$\nabla\cdot\mathbf{A} = \frac{1}{r}\frac{\partial}{\partial r}(rA_r) + \frac{1}{r}\frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

$$\nabla\times\mathbf{A} = \frac{1}{r}\begin{vmatrix} \hat{\mathbf{r}} & \hat{\boldsymbol{\phi}}r & \hat{\mathbf{z}} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_r & rA_\phi & A_z \end{vmatrix}$$

$$= \hat{\mathbf{r}}\left(\frac{1}{r}\frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z}\right) + \hat{\boldsymbol{\phi}}\left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r}\right) + \hat{\mathbf{z}}\frac{1}{r}\left[\frac{\partial}{\partial r}(rA_\phi) - \frac{\partial A_r}{\partial \phi}\right]$$

$$\nabla^2 V = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

常用导数

$f(x)$	$\frac{df}{dx}$
x^a	ax^{a-1}
$\frac{1}{x^a}$	$-ax^{-(a+1)}$
$\sqrt{x}$	$\frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$
e^{ax}	ae^{ax}
b^x	$b^x \ln(b)$
$\ln(x)$	$\frac{1}{x}$
$\log_a(x)$	$\frac{1}{x\ln(a)}$
$\sin(ax)$	$a\cos(ax)$
$\cos(ax)$	$-a\sin(ax)$
$\tan(ax)$	$a\sec^2(ax)$

常用积分

$$\int x^n\,dx = \frac{1}{n+1}x^{n+1} + C$$

$$\int \frac{1}{x}\,dx = \ln|x| + C$$

$$\int x^{-n}\,dx = \frac{1}{-n+1}x^{-n+1} + C$$

$$\int \frac{1}{ax+b}\,dx = \frac{1}{a}\ln|ax+b| + C$$

$$\int \cos(ax)\,dx = \frac{1}{a}\sin(ax) + C$$

$$\int \sin(ax)\,dx = -\frac{1}{a}\cos(ax) + C$$

$$\int \tan(ax)\,dx = \frac{1}{a}\ln|\sec(ax)| + C$$

$$\int e^{ax}\,dx = \frac{1}{a}e^{ax} + C$$

$$\int b^x\,dx = \frac{b^x}{\ln(b)} + C$$

$$\int \ln x\,dx = x\,\ln(x) - x + C$$

$$\int e^{ax}\sin(bx)\,dx = \frac{e^{ax}}{a^2+b^2}(a\sin(bx) - b\cos(bx)) + C$$

$$\int e^{ax}\cos(bx)\,dx = \frac{e^{ax}}{a^2+b^2}(a\cos(bx) + b\sin(bx)) + C$$

$$\int xe^x\,dx = (x-1)e^x + C$$

梯度、散度、旋度与拉普拉斯算子

直角坐标（x, y, z）

$$\nabla V = \hat{\mathbf{x}}\frac{\partial V}{\partial x} + \hat{\mathbf{y}}\frac{\partial V}{\partial y} + \hat{\mathbf{z}}\frac{\partial V}{\partial z}$$

$$\nabla\cdot\mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla\times\mathbf{A} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$= \hat{\mathbf{x}}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \hat{\mathbf{y}}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \hat{\mathbf{z}}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

圆柱坐标（r, φ, z）

$$\nabla V = \hat{\mathbf{r}}\frac{\partial V}{\partial r} + \hat{\boldsymbol{\phi}}\frac{1}{r}\frac{\partial V}{\partial \phi} + \hat{\mathbf{z}}\frac{\partial V}{\partial z}$$

$$\nabla\cdot\mathbf{A} = \frac{1}{r}\frac{\partial}{\partial r}(rA_r) + \frac{1}{r}\frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

$$\nabla\times\mathbf{A} = \frac{1}{r}\begin{vmatrix} \hat{\mathbf{r}} & \hat{\boldsymbol{\phi}}r & \hat{\mathbf{z}} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_r & rA_\phi & A_z \end{vmatrix}$$

$$= \hat{\mathbf{r}}\left(\frac{1}{r}\frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z}\right) + \hat{\boldsymbol{\phi}}\left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r}\right) + \hat{\mathbf{z}}\frac{1}{r}\left[\frac{\partial}{\partial r}(rA_\phi) - \frac{\partial A_r}{\partial \phi}\right]$$

$$\nabla^2 V = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

Spherical Coordinates ($R,\ \theta,\ \phi$)

$$\nabla V = \hat{\mathbf{R}}\frac{\partial V}{\partial R} + \hat{\boldsymbol{\theta}}\frac{1}{R}\frac{\partial V}{\partial \theta} + \hat{\boldsymbol{\phi}}\frac{1}{R\sin\theta}\frac{\partial V}{\partial \phi}$$

$$\nabla\cdot\mathbf{A} = \frac{1}{R^2}\frac{\partial}{\partial R}(R^2 A_R) + \frac{1}{R\sin\theta}\frac{\partial}{\partial\theta}(A_\theta\sin\theta) + \frac{1}{R\sin\theta}\frac{\partial A_\phi}{\partial\phi}$$

$$\nabla\times\mathbf{A} = \frac{1}{R^2\sin\theta}\begin{vmatrix} \hat{\mathbf{R}} & \hat{\boldsymbol{\theta}}R & \hat{\boldsymbol{\phi}}R\sin\theta \\ \dfrac{\partial}{\partial R} & \dfrac{\partial}{\partial\theta} & \dfrac{\partial}{\partial\phi} \\ A_R & RA_\theta & (R\sin\theta)A_\phi \end{vmatrix}$$

$$= \hat{\mathbf{R}}\frac{1}{R\sin\theta}\left[\frac{\partial}{\partial\theta}(A_\phi\sin\theta) - \frac{\partial A_\theta}{\partial\phi}\right] + \hat{\boldsymbol{\theta}}\frac{1}{R}\left[\frac{1}{\sin\theta}\frac{\partial A_R}{\partial\phi} - \frac{\partial}{\partial R}(RA_\phi)\right] + \hat{\boldsymbol{\phi}}\frac{1}{R}\left[\frac{\partial}{\partial R}(RA_\theta) - \frac{\partial A_R}{\partial\theta}\right]$$

$$\nabla^2 V = \frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial V}{\partial R}\right) + \frac{1}{R^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial V}{\partial\theta}\right) + \frac{1}{R^2\sin^2\theta}\frac{\partial^2 V}{\partial\phi^2}$$

Some Useful Vector Identities

$\mathbf{A}\cdot\mathbf{B} = AB\cos\theta_{AB}$ Scalar (or dot) product

$\mathbf{A}\times\mathbf{B} = \hat{\mathbf{n}}AB\sin\theta_{AB}$ Vector (or cross) product, $\hat{\mathbf{n}}$ normal to plane containing $\mathbf{A}$ and $\mathbf{B}$

$\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = \mathbf{B}\cdot(\mathbf{C}\times\mathbf{A}) = \mathbf{C}\cdot(\mathbf{A}\times\mathbf{B})$

$\mathbf{A}\times(\mathbf{B}\times\mathbf{C}) = \mathbf{B}(\mathbf{A}\cdot\mathbf{C}) - \mathbf{C}(\mathbf{A}\cdot\mathbf{B})$

$\nabla(U+V) = \nabla U + \nabla V$

$\nabla(UV) = U\nabla V + V\nabla U$

$\nabla\cdot(\mathbf{A}+\mathbf{B}) = \nabla\cdot\mathbf{A} + \nabla\cdot\mathbf{B}$

$\nabla\cdot(U\mathbf{A}) = U\nabla\cdot\mathbf{A} + \mathbf{A}\cdot\nabla U$

$\nabla\times(U\mathbf{A}) = U\nabla\times\mathbf{A} + \nabla U\times\mathbf{A}$

$\nabla\times(\mathbf{A}+\mathbf{B}) = \nabla\times\mathbf{A} + \nabla\times\mathbf{B}$

$\nabla\cdot(\mathbf{A}\times\mathbf{B}) = \mathbf{B}\cdot(\nabla\times\mathbf{A}) - \mathbf{A}\cdot(\nabla\times\mathbf{B})$

$\nabla\cdot(\nabla\times\mathbf{A}) = 0$

$\nabla\times\nabla V = 0$

$\nabla\cdot\nabla V = \nabla^2 V$

$\nabla\times\nabla\times\mathbf{A} = \nabla(\nabla\cdot\mathbf{A}) - \nabla^2\mathbf{A}$

$\displaystyle\int_v (\nabla\cdot\mathbf{A})\,d\mathcal{v} = \oint_S \mathbf{A}\cdot d\mathbf{s}$ Divergence theorem (S encloses $\mathcal{v}$)

$\displaystyle\int_S (\nabla\times\mathbf{A})\cdot d\mathbf{s} = \oint_C \mathbf{A}\cdot d\mathbf{l}$ Stokes's theorem (S bounded by C)

球坐标（R, θ, φ）

$$\nabla V = \hat{\mathbf{R}}\frac{\partial V}{\partial R} + \hat{\boldsymbol{\theta}}\frac{1}{R}\frac{\partial V}{\partial \theta} + \hat{\boldsymbol{\phi}}\frac{1}{R\sin\theta}\frac{\partial V}{\partial \phi}$$

$$\nabla\cdot\mathbf{A} = \frac{1}{R^2}\frac{\partial}{\partial R}(R^2 A_R) + \frac{1}{R\sin\theta}\frac{\partial}{\partial\theta}(A_\theta\sin\theta) + \frac{1}{R\sin\theta}\frac{\partial A_\phi}{\partial\phi}$$

$$\nabla\times\mathbf{A} = \frac{1}{R^2\sin\theta}\begin{vmatrix} \hat{\mathbf{R}} & \hat{\boldsymbol{\theta}}R & \hat{\boldsymbol{\phi}}R\sin\theta \\ \dfrac{\partial}{\partial R} & \dfrac{\partial}{\partial\theta} & \dfrac{\partial}{\partial\phi} \\ A_R & RA_\theta & (R\sin\theta)A_\phi \end{vmatrix}$$

$$= \hat{\mathbf{R}}\frac{1}{R\sin\theta}\left[\frac{\partial}{\partial\theta}(A_\phi\sin\theta) - \frac{\partial A_\theta}{\partial\phi}\right] + \hat{\boldsymbol{\theta}}\frac{1}{R}\left[\frac{1}{\sin\theta}\frac{\partial A_R}{\partial\phi} - \frac{\partial}{\partial R}(RA_\phi)\right] + \hat{\boldsymbol{\phi}}\frac{1}{R}\left[\frac{\partial}{\partial R}(RA_\theta) - \frac{\partial A_R}{\partial\theta}\right]$$

$$\nabla^2 V = \frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial V}{\partial R}\right) + \frac{1}{R^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial V}{\partial\theta}\right) + \frac{1}{R^2\sin^2\theta}\frac{\partial^2 V}{\partial\phi^2}$$

常用矢量恒等式

$\mathbf{A}\cdot\mathbf{B} = AB\cos\theta_{AB}$ 标量积（点积）

$\mathbf{A}\times\mathbf{B} = \hat{\mathbf{n}}AB\sin\theta_{AB}$ 矢量积（叉积）
单位法向量垂直于 A、B 所在的平面。

$\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = \mathbf{B}\cdot(\mathbf{C}\times\mathbf{A}) = \mathbf{C}\cdot(\mathbf{A}\times\mathbf{B})$

$\mathbf{A}\times(\mathbf{B}\times\mathbf{C}) = \mathbf{B}(\mathbf{A}\cdot\mathbf{C}) - \mathbf{C}(\mathbf{A}\cdot\mathbf{B})$

$\nabla(U+V) = \nabla U + \nabla V$

$\nabla(UV) = U\nabla V + V\nabla U$

$\nabla\cdot(\mathbf{A}+\mathbf{B}) = \nabla\cdot\mathbf{A} + \nabla\cdot\mathbf{B}$

$\nabla\cdot(U\mathbf{A}) = U\nabla\cdot\mathbf{A} + \mathbf{A}\cdot\nabla U$

$\nabla\times(U\mathbf{A}) = U\nabla\times\mathbf{A} + \nabla U\times\mathbf{A}$

$\nabla\times(\mathbf{A}+\mathbf{B}) = \nabla\times\mathbf{A} + \nabla\times\mathbf{B}$

$\nabla\cdot(\mathbf{A}\times\mathbf{B}) = \mathbf{B}\cdot(\nabla\times\mathbf{A}) - \mathbf{A}\cdot(\nabla\times\mathbf{B})$

$\nabla\cdot(\nabla\times\mathbf{A}) = 0$

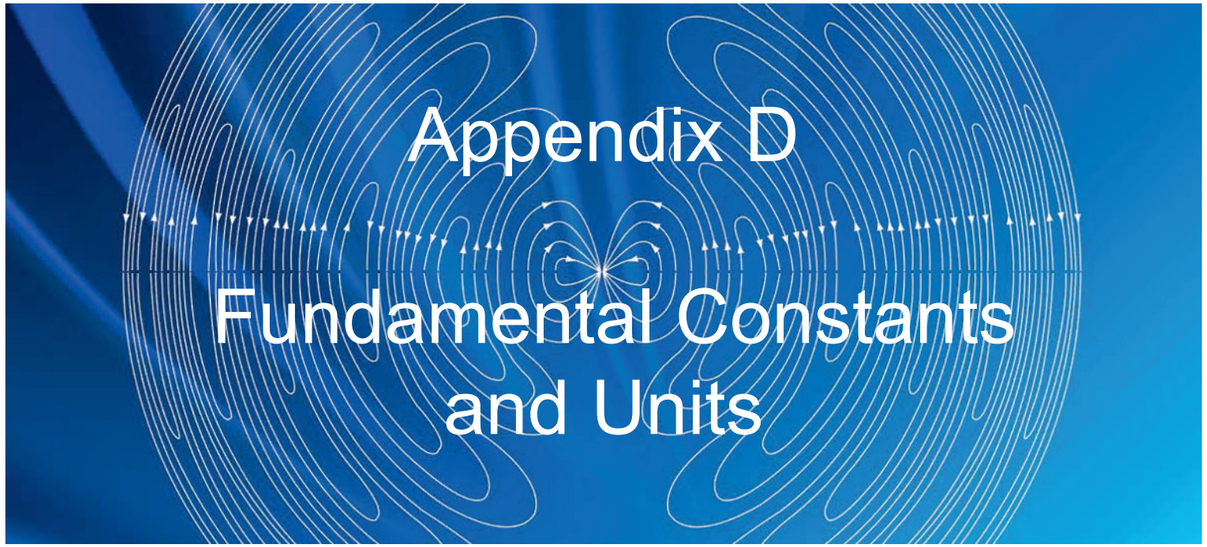
$\nabla\times\nabla V = 0$

$\nabla\cdot\nabla V = \nabla^2 V$

$\nabla\times\nabla\times\mathbf{A} = \nabla(\nabla\cdot\mathbf{A}) - \nabla^2\mathbf{A}$

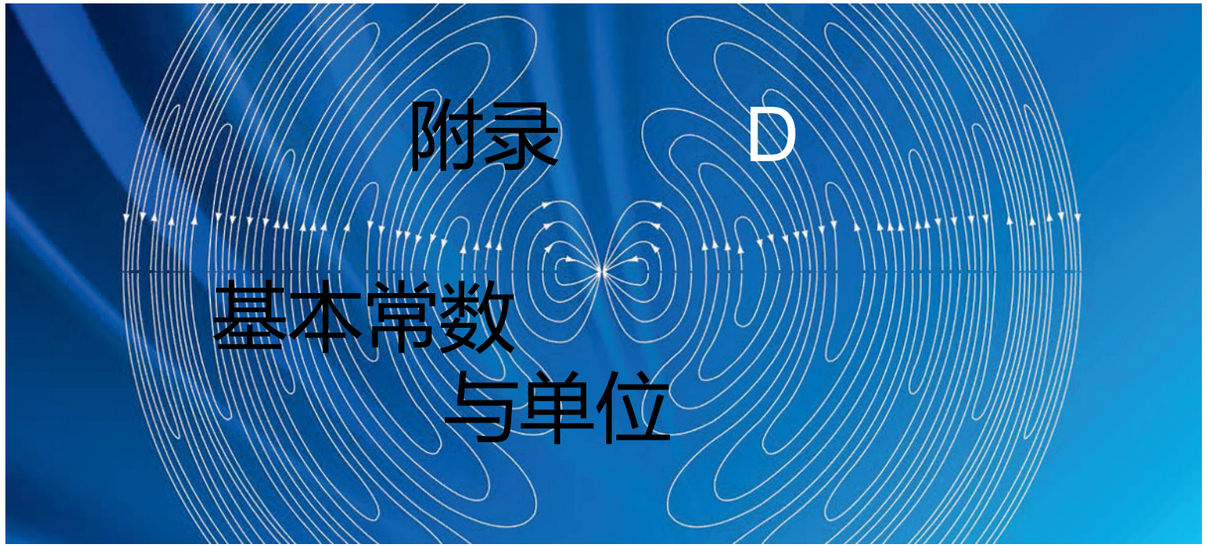
$\displaystyle\int_v (\nabla\cdot\mathbf{A})\,d\mathcal{v} = \oint_S \mathbf{A}\cdot d\mathbf{s}$ 散度定理
（S 包围体积 v）

$\displaystyle\int_S (\nabla\times\mathbf{A})\cdot d\mathbf{s} = \oint_C \mathbf{A}\cdot d\mathbf{l}$ 斯托克斯定理
（S 以 C 为边界）



Appendix D

Fundamental Constants and Units



附录 D

基本常数与单位

FUNDAMENTAL PHYSICAL CONSTANTS

CONSTANT	SYMBOL	VALUE
Speed of light in vacuum	c	$2.998 \times 10^{8} \approx 3 \times 10^{8}$ m/s
Gravitational constant	G	6.67×10^{-11} N·m^2/kg^2
Boltzmann's constant	K	1.38×10^{-23} J/K
Elementary charge	e	1.60×10^{-19} C
Permittivity of free space	ε_0	$8.85 \times 10^{-12} \approx \frac{1}{36\pi} \times 10^{-9}$ F/m
Permeability of free space	μ_0	$4\pi \times 10^{-7}$ H/m
Electron mass	m_e	9.11×10^{-31} kg
Proton mass	m_p	1.67×10^{-27} kg
Planck's constant	h	6.63×10^{-34} J·s
Intrinsic impedance of free space	η_0	$376.7 \approx 120\pi\ \Omega$

基本物理常数

常数	符号	数值
真空中的光速	c	$2.998 \times 10^{8} \approx 3 \times 10^{8}$ m/s
万有引力常数	G	6.67×10^{-11} N·m^2/kg^2
玻尔兹曼常数	K	1.38×10^{-23} J/K
元电荷	e	1.60×10^{-19} C
真空介电常数	ε_0	$8.85 \times 10^{-12} \approx \frac{1}{36\pi} \times 10^{-9}$ F/m
真空磁导率	μ_0	$4\pi \times 10^{-7}$ H/m
电子质量	m_e	9.11×10^{-31} kg
质子质量	m_p	1.67×10^{-27} kg
普朗克常数	h	6.63×10^{-34} J·s
真空本征阻抗	η_0	$376.7 \approx 120\pi\ \Omega$